Mathematics practice booklet: Grade 2 expectations

Grade 2 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 2 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: 2.OA.A.1, 2.OA.B.2, 2.OA.C.3, 2.OA.C.4, 2.MD.B.6

A worked example

Maryam has 27 stickers. Her cousin gives her 15 more. How many stickers does she have now?

1. Add the tens first: 27 + 10 = 37.
2. Then add the ones: 37 + 5 = 42.

42 stickers

Questions

1. Add: 8 + 7

   A. 14

   B. 15

   C. 16

2. Hamdan has 14 AED and buys a juice for 6 AED. How many dirhams are left?

   _____

3. Find the missing number: 9 + __ = 17

   _____

4. Salem has 56 dates. He eats 23 of them. How many dates are left?

   _____

5. Which number is even?

   A. 19

   B. 25

   C. 32

6. Maryam and her cousin Hind share 16 date cookies equally with none left over. How many cookies does each get? Write 16 as a sum of two equal numbers.

   _____

7. There are 5 plates with 2 date cookies on each plate. Write an addition sentence and find the total.

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8. A tray of fruit has 3 rows with 4 apples in each row. Which addition sentence gives the number of apples?

   A. 4 + 4 + 4 = 12

   B. 3 + 4 = 7

   C. 4 + 4 + 4 + 4 = 16

9. Khalid has 28 marbles. His uncle's family gives him 15 more and then he gives 9 to his brother. How many marbles does he have now?

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10. Layla and Noura want to share 17 sweets equally. What happens?

    A. Each gets 9 and none are left

    B. Each gets 8 and 1 is left over

    C. Each gets 8 and none are left

11. Add: 6 + 4 + 9. Which two numbers make a ten?

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12. Omar starts at 8 on a number line and jumps 5 more. Where does he land?

    A. 12

    B. 14

    C. 13

Section 2: Shapes, measuring and space

Standards practised: 2.G.A.1, 2.G.A.2, 2.G.A.3, 2.MD.A.1, 2.MD.A.2, 2.MD.A.3, 2.MD.A.4, 2.MD.B.5, 2.MD.C.7, 2.MD.C.8

A worked example

The short hand of a clock points just past the 3 and the long hand points to the 5. What time is it?

1. The short hand shows the hour: 3.
2. The long hand on the 5 means 5 × 5 = 25 minutes.

3:25

Questions

1. How many sides and corners does a triangle have?

   A. 3 sides and 3 corners

   B. 3 sides and 4 corners

   C. 4 sides and 4 corners

2. Maha covers a wall with square tiles: 2 rows with 3 equal tiles in each row. How many tiles is that? Write an addition sentence.

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3. How many flat faces does a cube-shaped box have?

   A. 4

   B. 6

   C. 8

4. Noura cuts a rectangular sandwich into 2 equal parts. What is each part called?

   A. A quarter

   B. A third

   C. A half

5. A square pizza is cut into 4 equal parts. What is each part called? Do 4 parts make a whole?

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6. A ribbon is 15 cm long. Another ribbon is 9 cm long. How much longer is the first ribbon?

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7. Which is the best estimate for the length of a pencil?

   A. 150 centimetres

   B. 15 metres

   C. 15 centimetres

8. The long hand points to the 9 and the short hand is just before the 8. What time is it?

   A. 7:45

   B. 8:45

   C. 7:09

9. The school bus leaves at 8:55 in the morning. The bell rings 5 minutes later. What time does the bell ring? Is it a.m. or p.m.?

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10. Layla buys a notebook for 6 AED and a pencil for 3 AED. She pays with a 10 AED note. How much change does she get?

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11. Omar has one 10 AED note, one 5 AED note and three 1 AED coins. How much money does he have?

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12. Maryam measures a table with a 20 cm ruler and gets 4 rulers. Then she measures it with a 10 cm strip and gets 8 strips. Why are the two numbers different?

Section 3: Data, graphs and chance

Standards practised: 2.MD.D.9, 2.MD.D.10

A worked example

A line plot shows the lengths of leaves in cm. There are 2 dots above 3, 3 dots above 4 and 1 dot above 5. How many leaves were measured?

1. Each dot is one leaf.
2. Count the dots: 2 + 3 + 1 = 6.

6 leaves

Questions

1. Children chose their favourite fruit: dates 7, mangoes 5, bananas 4, apples 6. How many children chose dates?

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2. In the same table, which fruit did the most children choose?

   A. Mangoes

   B. Apples

   C. Dates

3. How many more children chose dates than bananas?

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4. How many children chose a fruit in all?

   A. 22

   B. 18

   C. 21

5. How many children chose mangoes or apples?

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6. In a picture graph of favourite drinks, each smiley stands for 1 child. Karak has 9 smileys. How many children chose karak?

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7. In a bar graph, Salem's bar goes up to 8 books and Hind's bar goes up to 5 books. How many more books did Salem read?

   A. 13

   B. 3

   C. 4

8. Use the leaf line plot from the example (2 dots above 3, 3 dots above 4, 1 dot above 5). How many leaves are 4 cm long?

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9. In the leaf line plot (2 dots above 3, 3 dots above 4, 1 dot above 5), how long is the longest leaf?

   A. 5 cm

   B. 3 cm

   C. 4 cm

10. Noura sells date cookies and counts them with tally marks: 2 groups of five and 3 more lines. How many cookies did she sell?

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11. In the fruit table, 4 more children choose bananas. Which fruit is now the most popular?

Section 4: Numbers and calculating

Standards practised: 2.NBT.A.1, 2.NBT.A.2, 2.NBT.A.3, 2.NBT.A.4, 2.NBT.B.5, 2.NBT.B.6, 2.NBT.B.7, 2.NBT.B.8, 2.NBT.B.9

A worked example

What is the value of each digit in 374? Write it in expanded form.

1. The 3 is in the hundreds place: 300.
2. The 7 is in the tens place: 70. The 4 is in the ones place: 4.

374 = 300 + 70 + 4

Questions

1. Write the number: 5 hundreds, 3 tens and 7 ones.

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2. Which is 482 in expanded form?

   A. 400 + 80 + 2

   B. 40 + 80 + 2

   C. 400 + 8 + 2

3. What is the value of the digit 6 in 361?

   A. 6

   B. 60

   C. 600

4. Salem has 408 date stones and Hamdan has 480. Which is true?

   A. 408 > 480

   B. 408 = 480

   C. 408 < 480

5. Three boxes hold 325, 235 and 253 dates. Put the numbers in order from least to greatest.

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6. Skip count by 5s: 35, 40, 45, __, __. What are the next two numbers?

   A. 50, 60

   B. 46, 47

   C. 50, 55

7. Skip count by 10s: 120, 130, 140, __, __

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8. Skip count by 100s: 300, 400, __, 600

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9. Add: 45 + 38

   A. 83

   B. 73

   C. 93

10. Mona has 74 stickers and gives 29 of them to her relatives. How many stickers does she have left?

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11. A shop has 357 dates in one box and 128 in another box. How many dates are there in all?

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12. What is 10 more than 395? What is 100 less than 250? Hamdan says 1 hundred is the same as 10 tens. Is he right?

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. B. 15. Make a ten: 8 + 2 = 10, then 10 + 5 = 15. If your child chose A (14), they counted one too few; if they chose C (16), one too many.
2. 8 AED. Take 6 away from 14: 14 - 6 = 8, because 14 - 4 = 10 and 10 - 2 = 8.
3. 8. Think of the opposite: 17 - 9 = 8, so 9 + 8 = 17.
4. 33. Take away the tens, then the ones: 56 - 20 = 36, then 36 - 3 = 33.
5. C. 32. It makes 16 pairs with none left over, so it is even. If your child chose A (19) or B (25), they did not check the last digit, which must be 0, 2, 4, 6 or 8.
6. 8 cookies each. Split 16 into two equal groups: 8 + 8 = 16, so 16 is even.
7. 2 + 2 + 2 + 2 + 2 = 10 cookies. There are 5 plates with 2 on each, so add 2 five times.
8. A. 4 + 4 + 4 = 12. because there are 3 rows of 4. If your child chose B (3 + 4), they added the number of rows to the size of a row.
9. 34. First add: 28 + 15 = 43. Then take away 9: 43 - 9 = 34.
10. B. Each gets 8 and 1 is left over. because 8 + 8 = 16 and 17 is odd. If your child chose A, 9 each needs 18 sweets; if they chose C, 8 each uses only 16.
11. 19. The pair 6 and 4 makes 10, and then 10 + 9 = 19.
12. C. 13. Start at 8 and jump 5: 8 + 5 = 13. If your child chose A (12), they counted the start as a jump; if they chose B (14), they jumped one too many.

Section 2: Shapes, measuring and space

1. A. 3 sides and 3 corners. A triangle has 3 sides and 3 corners. If your child chose C (4 sides and 4 corners), they described a square or a rectangle.
2. 6 tiles. There are 2 rows of 3, so 3 + 3 = 6.
3. B. 6. A cube has 6 flat faces, like the sides of a dice. If your child chose C (8), they counted the corners instead of the faces.
4. C. A half. Two equal parts are called halves. If your child chose A (a quarter), that is the name for 4 equal parts; if they chose B (a third), it is the name for 3.
5. A quarter. Four equal parts are quarters, and 4 quarters put back together make 1 whole pizza.
6. 6 cm. To compare, subtract: 15 - 9 = 6, because 9 + 6 = 15.
7. C. 15 centimetres. about the length of a hand. If your child chose A (150 centimetres), that is about the height of a child; if they chose B (15 metres), it is about a classroom.
8. A. 7:45. The short hand is before the 8, so the hour is 7, and 9 × 5 = 45 minutes. If your child chose B (8:45), they read the hour as the nearer number, 8.
9. 9:00 a.m. Adding 5 minutes to 8:55 reaches the next hour, 9:00, and it is morning, so a.m.
10. 1 AED. Add the prices: 6 + 3 = 9. Then take that from 10: 10 - 9 = 1.
11. 18 AED. Add the note first, then the coins: 10 + 5 = 15, and 15 + 3 = 18.
12. The 10 cm strip is a smaller unit, so more are needed. 4 × 20 = 80 and 8 × 10 = 80, so both give 80 cm.

Section 3: Data, graphs and chance

1. 7. Find the row for dates in the table and read its number: 7 children.

2. C. Dates. Compare the numbers 7, 5, 4 and 6: 7 is the biggest. If your child chose B (Apples), they may have read the last number instead of comparing them all.
3. 3 more. To find how many more, subtract: 7 - 4 = 3.
4. A. 22. from 7 + 5 + 4 + 6 = 22. If your child chose B (18), they left out the 4 children who chose bananas.
5. 11. Mangoes are 5 and apples are 6, so add them: 5 + 6 = 11.
6. 9. Each smiley is one child, so 9 smileys means 9 children.
7. B. 3. "How many more" means subtract: 8 - 5 = 3. If your child chose A (13), they added 8 + 5 instead of finding the difference.
8. 3. Find the column above 4 on the line plot and count its dots: there are 3, so 3 leaves.
9. A. 5 cm. The longest leaf is the dot farthest to the right, above 5. If your child chose C (4 cm), they picked the column with the most dots, which is the most common length, not the longest.
10. 13. Count the groups of five first: 5 + 5 = 10. Then add the 3 extra lines: 10 + 3 = 13.
11. Bananas. With 4 more, bananas are 4 + 4 = 8, and 8 is more than the 7 for dates.

Section 4: Numbers and calculating

1. 537. Write the digits in order: 500 + 30 + 7 = 537.
2. A. 400 + 80 + 2. because 4 is in the hundreds place and 8 is in the tens place. If your child chose B (40 + 80 + 2), they treated the 4 as tens.
3. B. 60. The 6 is in the tens place, so it is worth 6 tens. If your child chose A (6), they used its face value; if they chose C (600), they put it in the hundreds place.
4. C. 408 < 480. Both have 4 hundreds, and 0 tens is less than 8 tens. If your child chose A (408 > 480), they may have compared the ones digits, 8 and 0.
5. 235, 253, 325. Compare the hundreds first. For the two with 2 hundreds, 3 tens is less than 5 tens, so 235 comes before 253.
6. C. 50, 55. Each jump adds 5: 45 + 5 = 50, then 50 + 5 = 55. If your child chose B (46, 47), they counted by ones; if they chose A (50, 60), they switched to tens.
7. 150, 160. Each number is 10 more than the one before: 140 + 10 = 150, then 150 + 10 = 160.
8. 500. Each step adds 100: 400 + 100 = 500, and 500 + 100 = 600.
9. A. 83. 5 + 8 = 13, so regroup the ten: 40 + 30 + 13 = 83. If your child chose B (73), they forgot the ten from 13; if they chose C (93), they added an extra ten.
10. 45. Take away 30, then add 1 back, because 29 is 1 less than 30: 74 - 30 = 44, then 44 + 1 = 45.
11. 485. Add by place: 300 + 100 = 400, 50 + 20 = 70, 7 + 8 = 15, and 400 + 70 + 15 = 485.
12. 405; 150; yes. 395 + 10 = 405 and 250 - 100 = 150. 10 tens count up to 100, so 10 tens make 1 hundred.